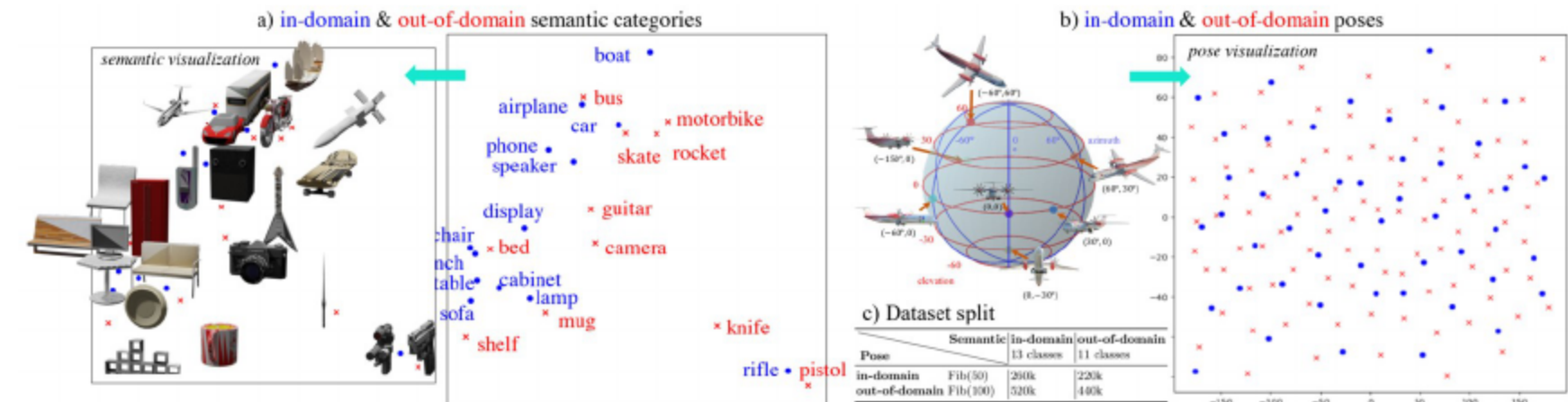


Pose-Aware SSL with Viewpoint Trajectory Regularization

Jiayun Wang; Yubei Chen; Stella X. Yu

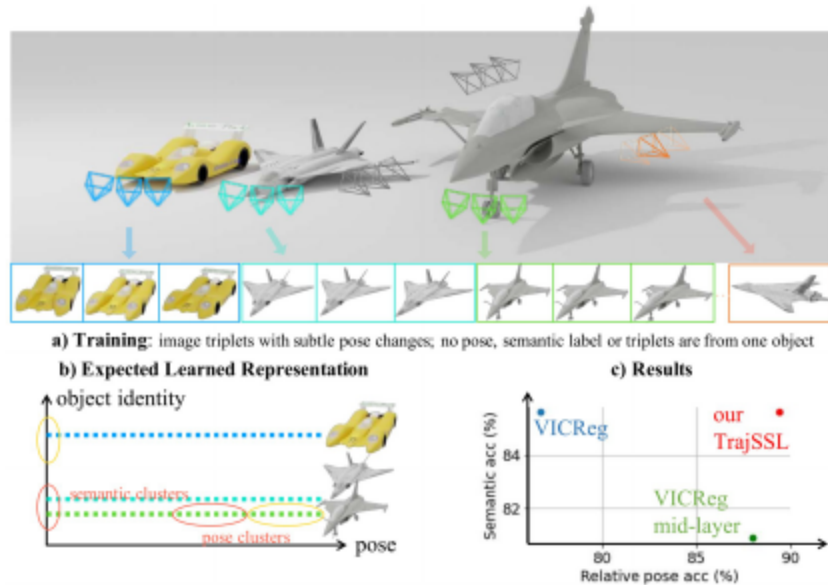
01 Why pose-aware SSL now?



- Invariant SSL underperforms on rare or unseen poses
- Lack of a standardized pose-aware SSL evaluation
- Motivation: safety-critical viewpoint reasoning and robust generalization

[[4]]

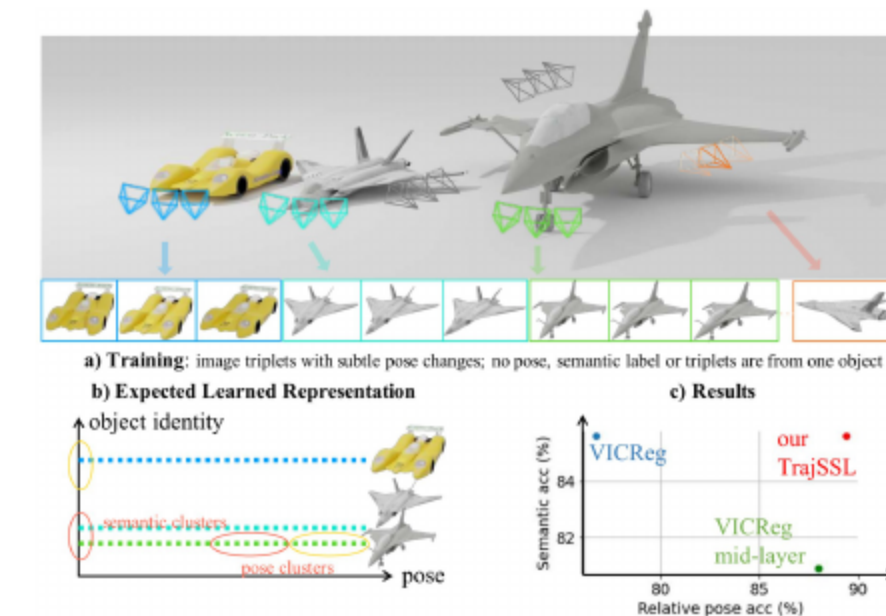
02 Goal: Joint identity + pose in SSL



- Jointly learn identity and pose from unlabeled triplets
 - Leverage adjacent views with small viewpoint changes
 - Infer global pose from local transitions
 - Preserve **semantic separability** while adding geometry
- Support both semantic classification and pose estimation
 - Maintain clear identity clusters
 - Enable accurate absolute and relative pose
 - Use task-appropriate readouts per layer
- Encourage smooth pose trajectories in feature space
 - Regularize adjacent views to align along

[[4]]

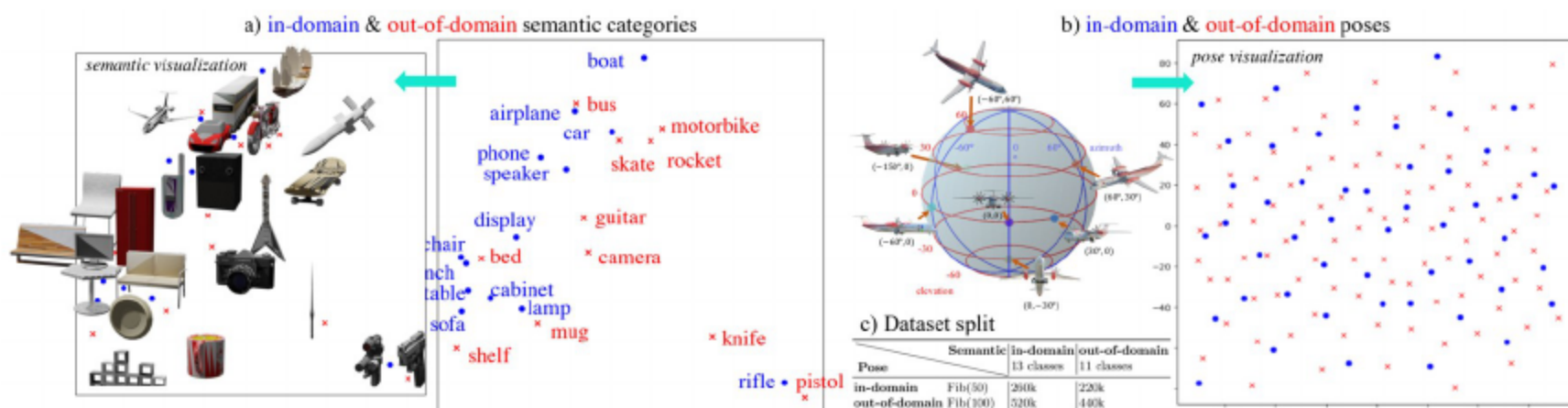
03 Invariant vs geometry-aware SSL



- Invariant SSL focuses on semantics and often shows **weak equivariance**
- Geometry-aware SSL seeks equivariance; our approach stays **label-free**
- Pose evaluation requires absolute and class-agnostic relative tests

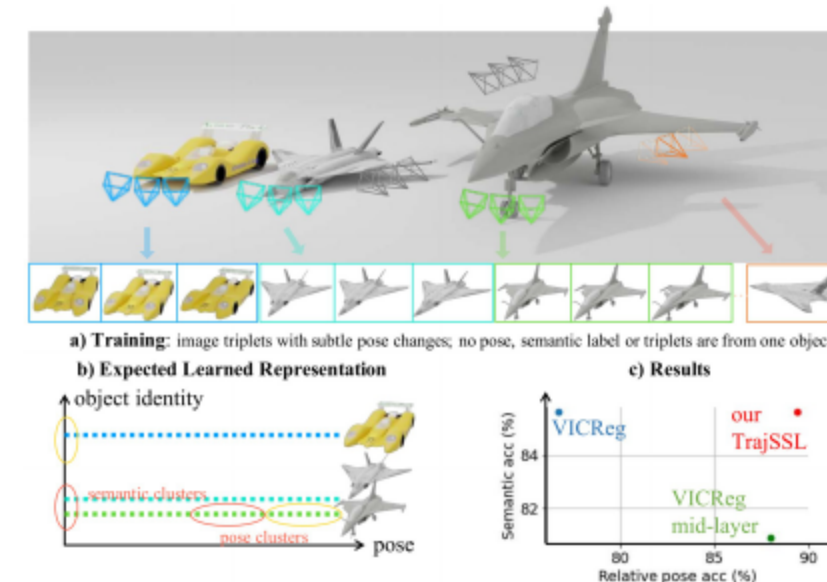
[Chen et al. 2003; Church et al. 2017]

04 Principles and scope



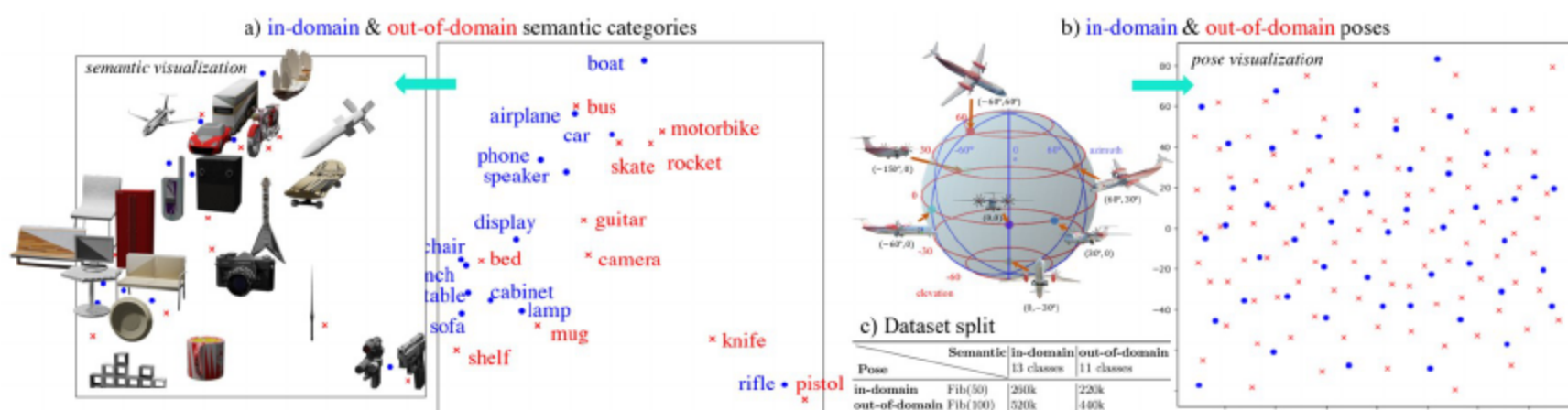
- Training uses images only with no semantic or pose labels
- Evaluation probes both semantics and geometry with consistent metrics
- Relative pose enables class-agnostic out-of-domain evaluation

05 Natural acquisition: adjacent triplets



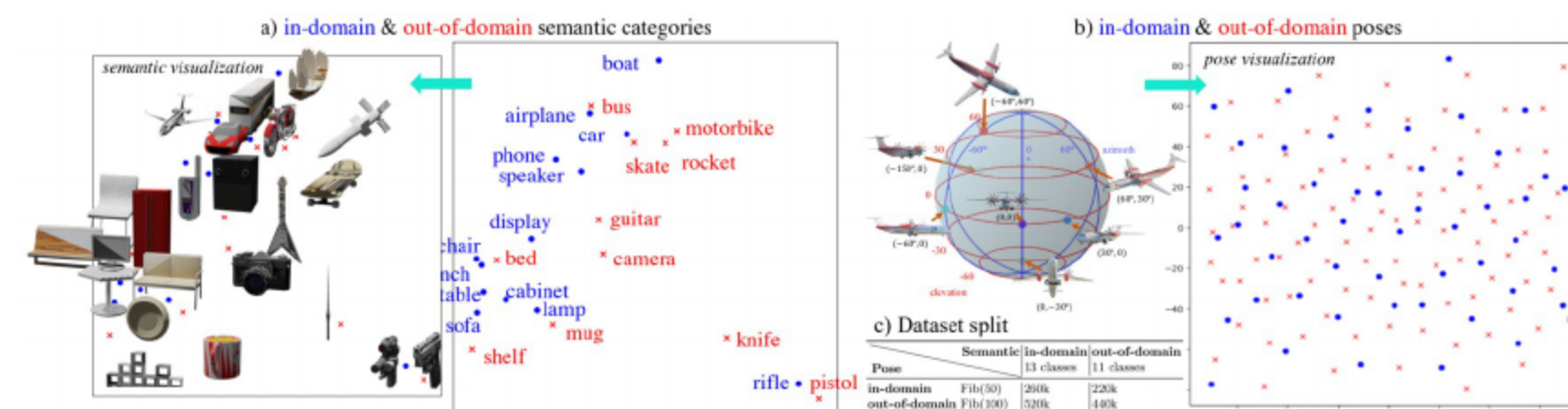
- Adjacent views arise naturally from smooth trajectories
 - Human micro-movements provide subtle pose shifts
 - Robot egomotion samples neighborhoods
 - Forms a **useful** self-supervision signal
- Synthetic adjacent triplets with small pose steps
 - Preserve geometric consistency
 - No identity or pose labels
 - Scalable generation procedure
- Trajectory-based regularization in SSL
 - Local linearity holds for small steps
 - Boosts **pose sensitivity** without labels
 - **Compatible with standard SSL**

06 Data generation and pose definitions



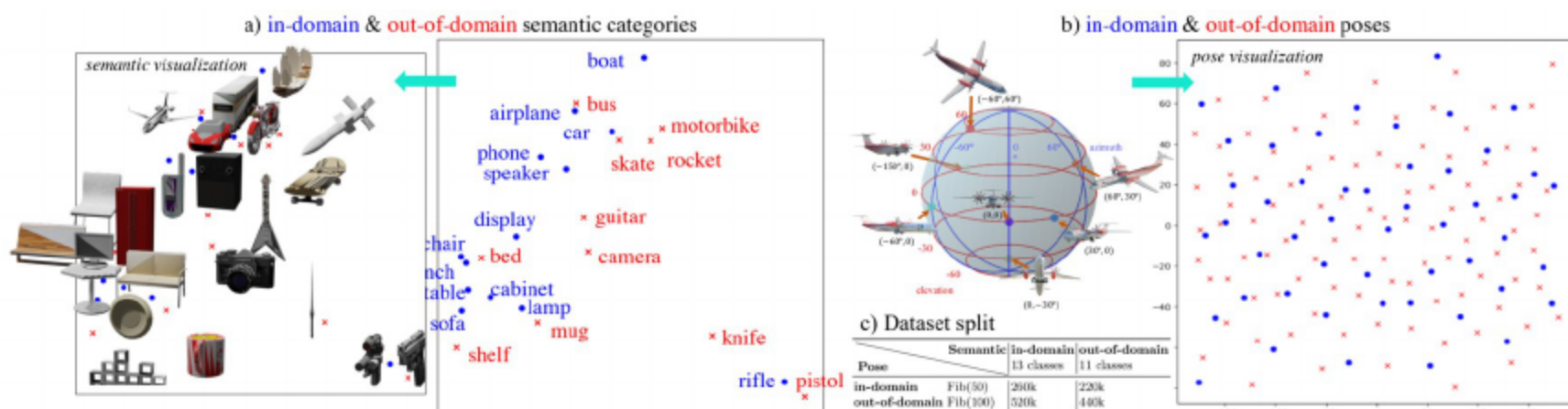
- ShapeNet renders with even viewpoint coverage and shadow-free images
- Pose defined by camera azimuth and elevation with consistent conventions
- Supports in-domain absolute pose and class-agnostic relative pose

07 Pose sampling and dataset splits



- Fibonacci spheres provide uniform viewpoints for training and testing
- Clear in-domain and out-of-domain splits across classes and poses
- Separation of train, probe, and test prevents **leakage**

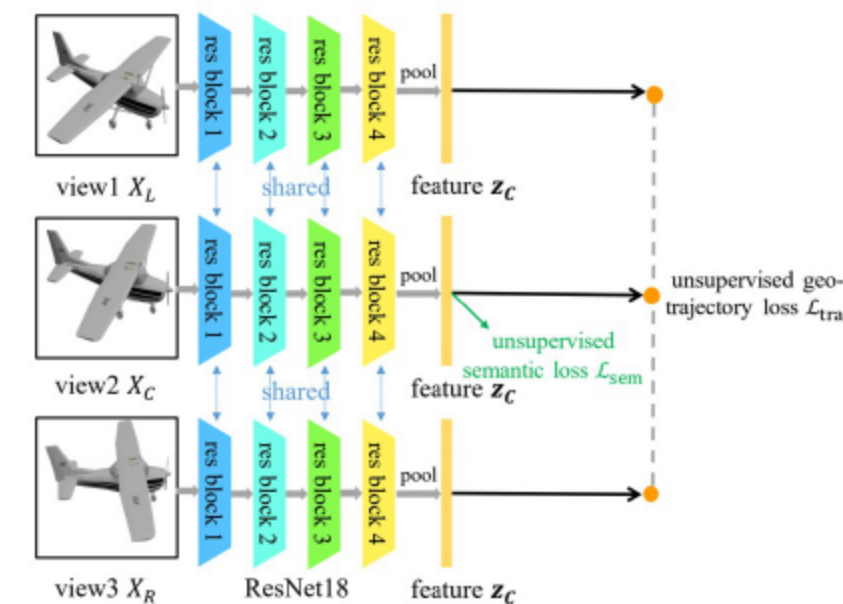
08 Downstream tasks and protocol



- In-domain: absolute with nearest neighbor; relative with a small probe
- Out-of-domain: unseen poses and unseen categories via relative pose
- Semantic baseline via linear probe with matched capacity

[Iwase et al. 2021]

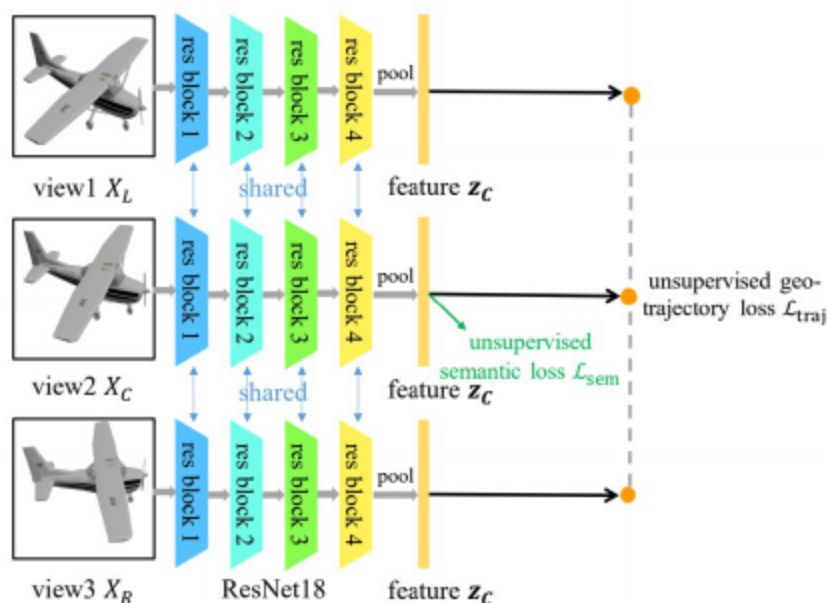
09 Baseline: invariant SSL representations



- Encoder with method-specific unsupervised semantic objectives
 - Two augmented views of the same image
 - Losses promote identity invariance
 - Standard data augmentations
- Pose information may persist in features
 - Absolute pose via nearest neighbor
 - Relative pose via a small probe
 - Treat pose as downstream signal
- Contrastive and non-contrastive variants
 - Avoid collapse while aligning positives
 - Strong class separation
 - Different regularizers

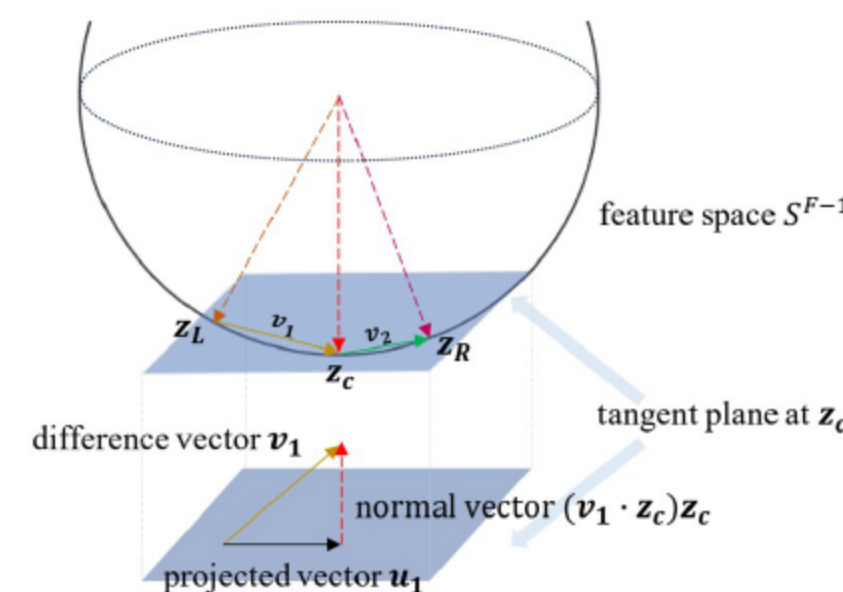
[[4]]

10 Trajectory-regularized SSL: intuition



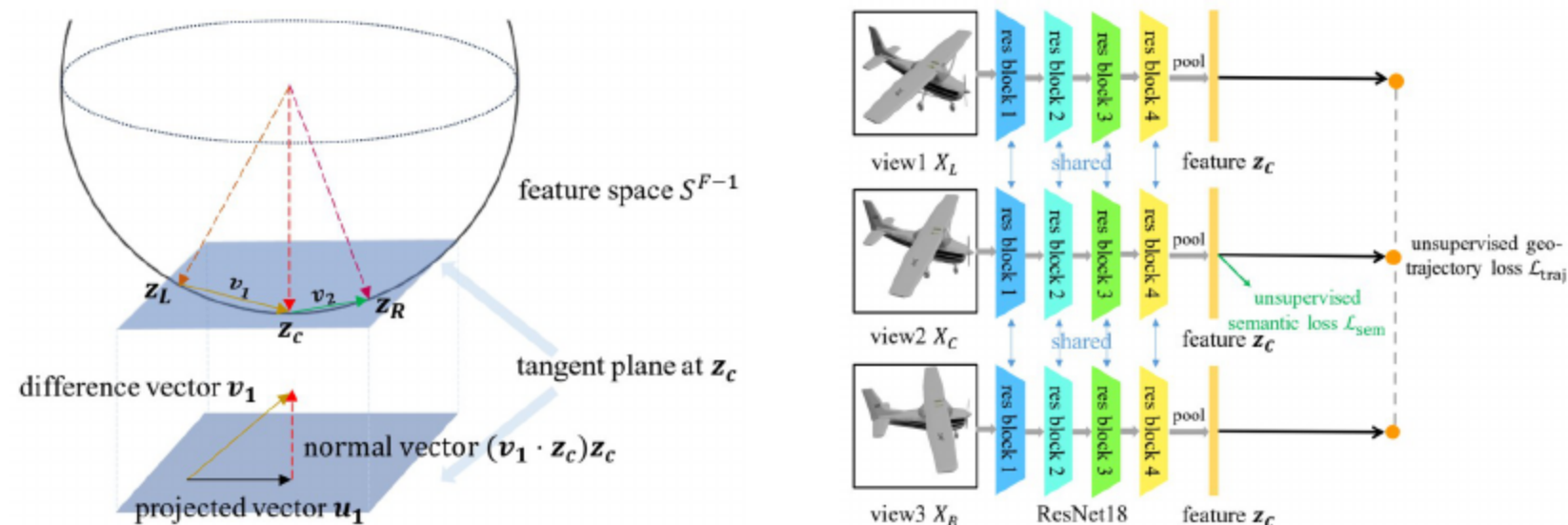
- Adjacent views should trace smooth, low-curvature paths
 - Enforce local linearity on normalized features
 - Align with geodesic structure for small pose steps
 - Explicitly encode **viewpoint change**
- Operate on pooled features without extra labels
 - Use triplets with subtle pose changes
 - Avoid random crops on triplets to prevent **geometry harm**
 - Standard augmentations applied elsewhere
- Complementary to invariant SSL
 - Add a trajectory loss alongside semantic

11 Trajectory loss on the hypersphere



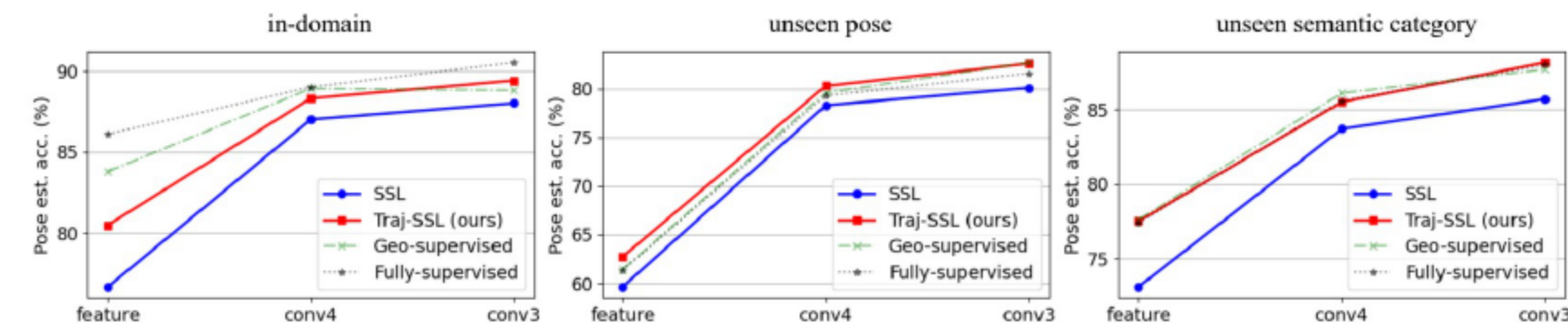
- Project adjacent-step differences into a tangent plane
 - Compute differences between neighboring features
 - Remove the normal component at the center point
 - Align steps within the tangent space
- Maximize cosine similarity between projected steps
 - Encourage local linearity in the tangent plane
 - Promote geodesic alignment across adjacent views
 - Stabilize **pose-aware structure**

12 Schematic + geodesic trajectory



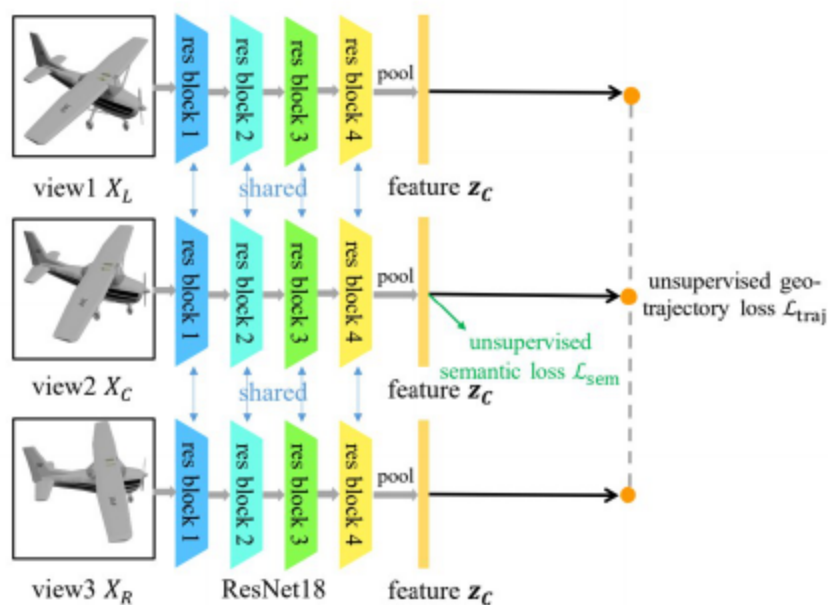
- Add a trajectory loss to standard SSL at pooled features
 - Semantic loss placement follows the baseline
 - Pose evaluation can use mid-layer representations
 - Single-encoder **multi-task** readiness
- Triplet-based local linear constraint encourages geodesics

13 Mid-layer features for pose: motivation



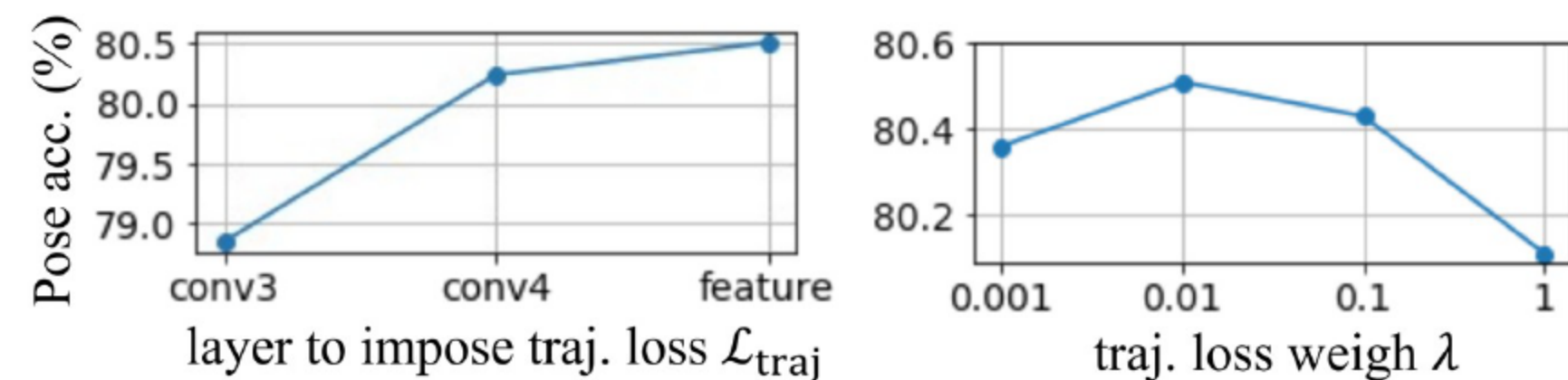
- Pose is mid-level; intermediate layers retain viewpoint cues
- Mid-layers outperform last layer for pose with minimal semantic impact
- Compression keeps most gains, enabling **efficient** deployment

14 Setup and baselines



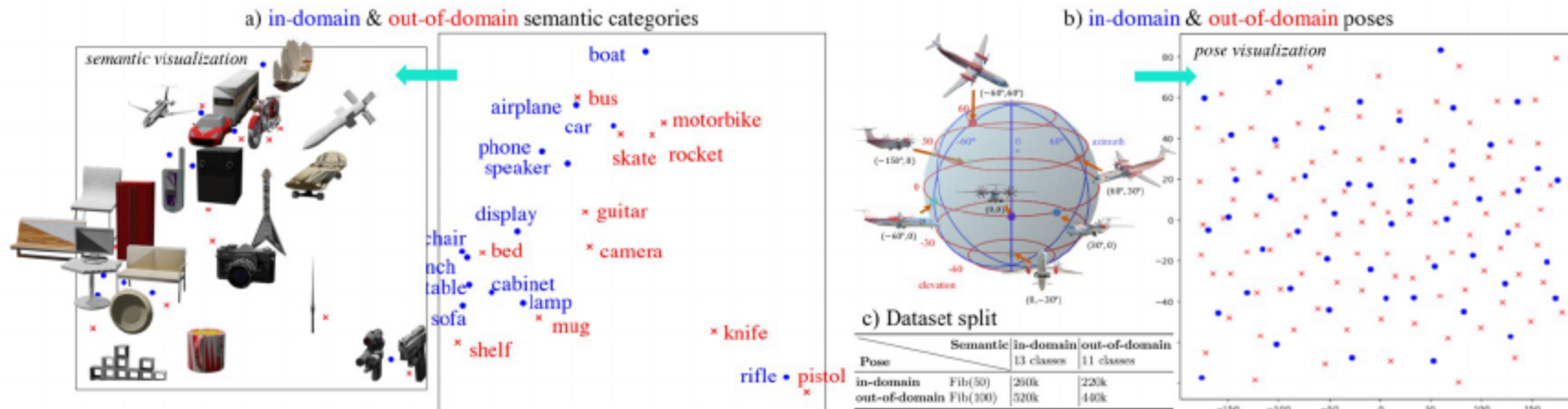
- Backbone and optimization
 - ResNet eighteen for all methods
 - Three hundred epochs with LARS and weight decay
 - Identical protocols across approaches
- Baselines: invariant SSL and supervised references
 - VICReg, SimCLR, SimSiam as SSL
 - Fully supervised and geometry-supervised references
 - Consistent data and augmentations
- Evaluation covers semantics and pose
 - Semantic via linear probe on frozen features
 - Absolute via nearest neighbor; relative via a small probe

15 Training specifics and analysis



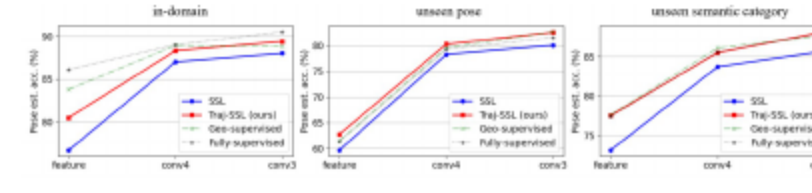
- Construct triplets by mirroring steps; avoid crops to prevent **distortion**
- Trajectory weight is small and not very sensitive; layer choice has minor effect
- Method is robust across backbones and non-equidistant poses

16 Main quantitative (feature z)



- Trajectory regularization boosts pose accuracy without hurting semantics
- Out-of-domain relative pose is competitive or slightly better than supervised
- Gains are consistent across VICReg, SimCLR, and SimSiam

17 Headline: mid-layer gains

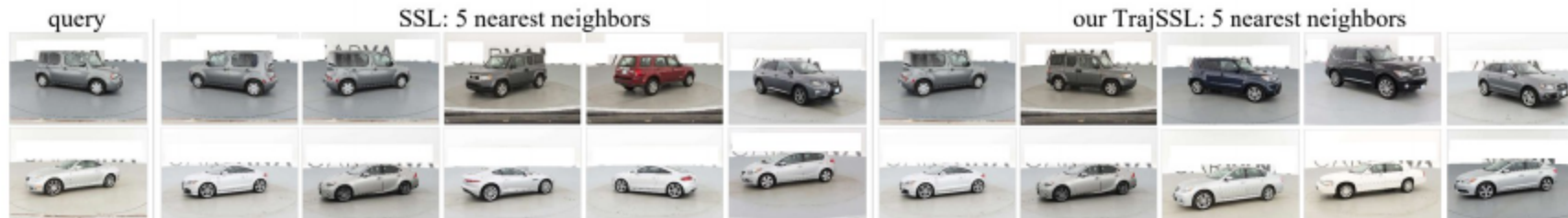


Pose Acc. (%)	In-Domain		Unseen Pose		Unseen Cat.	
Representation layer	conv3	conv4 feat	conv3	conv4 feat	conv3	conv4 feat
Fully-Sup.	90.5	89.0	86.1	81.6	79.3	61.3
Geometry-Sup.	88.8	88.9	83.8	82.7	79.6	61.4
VICReg [4]	88.0	87.0	76.7	80.1	78.3	59.6
VICReg+traj.	89.4(+1.4)	88.3	80.5	82.6(+2.5)	80.3	62.7

- Conv3 substantially improves relative pose, especially out-of-domain
 - Large gains on unseen poses and categories
 - Shrinks SSL to supervised gap
 - Best with matched probe size
- Keep last layer for semantics; mid-layer for

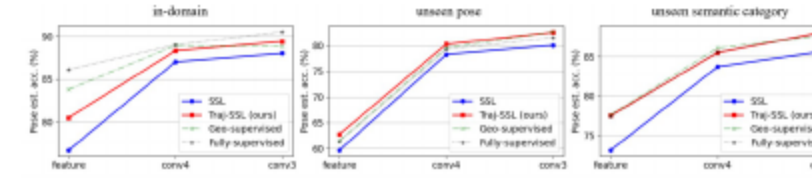
18

18 Transfer to real photos (Carvana)



- Synthetic-pretrained models transfer zero-shot to real data
- Trajectory regularization improves pose-aware retrieval
- SSL is competitive with supervised on real photos with robust results

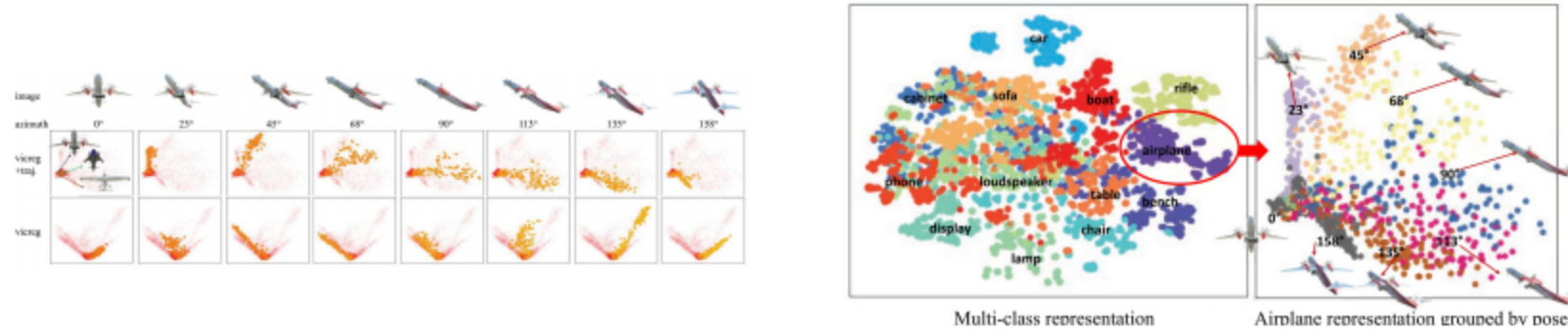
19 Mid-layer efficiency and compression



embedding	# dim	abs. pose acc. (%)	rel. pose acc. (%)
conv3	16,384	92.5	87.8
compressed conv3	512	91.4 (↓1.1)	82.4 (↓5.4)
conv4	8,192	91.9	85.2
compressed conv4	512	90.8 (↓1.1)	81.2 (↓4.0)
feature	512	87.8	77.5

- Mid-layer features are higher-dimensional but deliver better pose accuracy
 - Compute and memory increase modestly
 - Accuracy gains are substantial
 - Trade-offs are manageable
- Compression retains most gains and remains accurate

20



- Embeddings show semantic clusters with intra-class pose structure
 - Category-level grouping in pooled features
 - Within-class arcs by viewpoint
 - Joint organization is [clear](#)
- Trajectory regularization yields smoother pose manifolds



- Summary: trajectory loss plus mid-layer readouts improves pose with semantics preserved
- Limitations: synthetic-focused benchmark and limited geometric degrees of freedom
- Future: extend to full six degree pose, richer scenes, and larger backbones with [scalable](#) training

THANKS!